UH Mānoa Grid Forecasting

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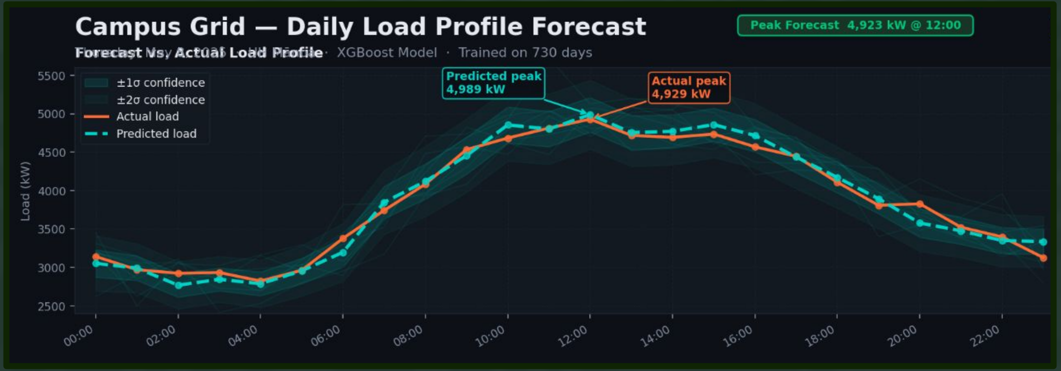
A campus electrical grid like UH Mānoa’s, could heavily benefit from knowing **future** electrical demand. This can help **avoid peak demand** charges, as well as help to manage **renewable energy generation.**

The Task

- Build a model that can forecast a daily load profile (24 Hr) using past ***meter & weather data.***



*sample visualization

Data Set:

- 35,000+ totalized meter readings (15 min intervals - 1 year)
- Open-Mateo Weather data (Hourly)

Approach

STAGE A — Tied-Weight Autoencoder

INPUT 13 features → ENCODER $\mathrm{ReLU}(Wx + b)$ → LATENT z 4 dims → DECODER $W^T z + b$ → $\hat{x}$ recon.

Single shared W · decoder uses W^T · one parameter, enforced by autograd

STAGE B — Feed-Forward Predictor (encoder frozen)

24h window (raw) → Encode each hour → Flatten 96 dims → FFNN $96 \to 64 \to 64 \to 24$ → $\hat{y}$ 24h kW

Approach

Latent Representations & Dimensionality Reduction

- Compresses 13 input features into a 4-dim latent code z

- More powerful than simple linear subspace projection due to retrieving more important weighted vector representing the entire weather

Tied Weights — W and W^T

- Encoder and decoder share one weight matrix W

- Decoder uses the transpose W^T — forces symmetric reconstruction

Encoder mapping

$$z = \sigma(W \cdot x + b\_enc)$$

$$z \in \mathbb{R}^4 \ (\text{latent code})$$

Encoder / Decoder

$$\text{Encoder: } z = \sigma(W \cdot x + b\_enc)$$

$$\text{Decoder: } \hat{x} = W^T \cdot z + b\_dec$$

Approach

MSE Loss Function & Connection to SVD

- Autoencoder trained with Mean Square Loss: minimizes $\|x - \hat{x}\|^2$

- ReLU breaks strict SVD equivalence — but buys nonlinear interactions PCA misses

Reconstruction Loss

$$L = {}^{1}/_{N} \sum \| x - \hat{x} \|^2$$

$$X = U \Sigma V^T \quad \textit{(SVD connection)}$$

Feed-Forward Predictor — Stage B

- Input: 96-dim flattened latent window (4 dims × 24 hours) instead of 312 raw features

- Encoder is frozen — Stage B only learns temporal dynamics, not feature representation

Predictor

$$z\_1, z\_2, \ldots, z\_24 \rightarrow \textit{flatten} \rightarrow \textit{FFNN}$$

$$\hat{y} = [\, \hat{y}_1, \hat{y}_2, \ldots, \hat{y}_{24} \,] \in \mathbb{R}^{24}$$

Approach

Ridge Regularization

- Prevents memorization by adding a penalty for relying on a single feature

$$\mathcal{L}_{\text{Ridge}} = \mathcal{L}_{\text{MSE}} + \lambda||\mathbf{W}||^2$$

Gradient Boosting (XGBoost)

- Temperature and campus load is non linear
- 300 decision trees for each hour

$$\hat{y} = \sum_{k=1}^{K} T_k(\mathbf{x}), \quad K = 300 \text{ trees per hour}$$

Approach

***Cooling Degree Hours**

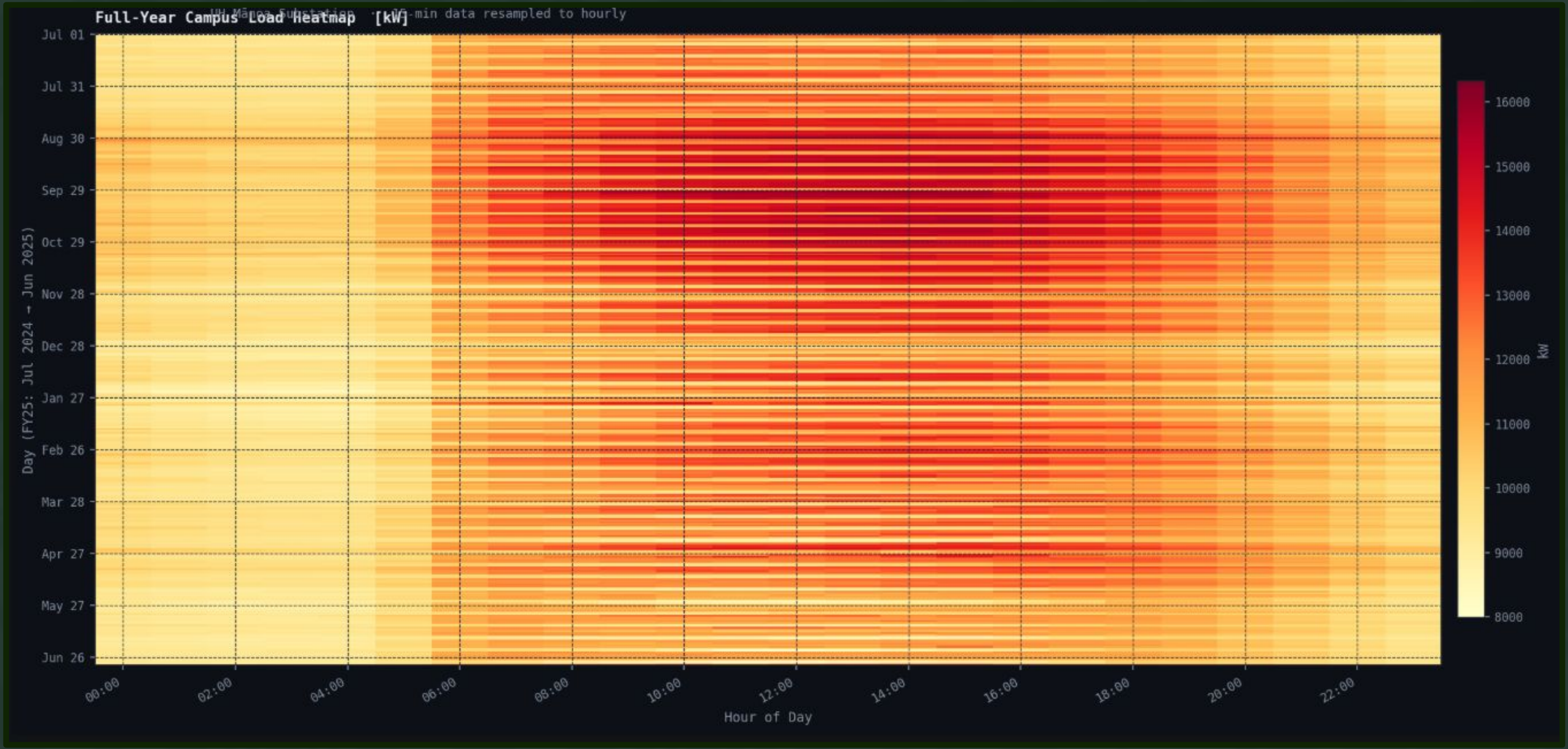
$$\mathrm{CDH}_h = \max(0,\ T_h - 65^\circ F)$$

When a building is 65° F it doesn't need air conditioning. This gives the model an accurate estimate of how much HVAC has been running throughout a day, which **greatly** impacts power consumption.


15-min meter CSV
Open-Meteo weather API
Merge on timestamp
8,760 hourly rows · 15 weather variables
Feature engineering
lag profiles · weather · calendar → 131 features
Temporal train / test split
297 training days → 60 held-out test days
XGBoost training
24 models × 300 trees, one per hour
Model evaluation
MAE 280 kW · MAPE 2.6% · profile r = 0.91
24-hour load forecast
predicted kW for each hour of tomorrow

Totalized Load Forecasting

Results



Heatmap

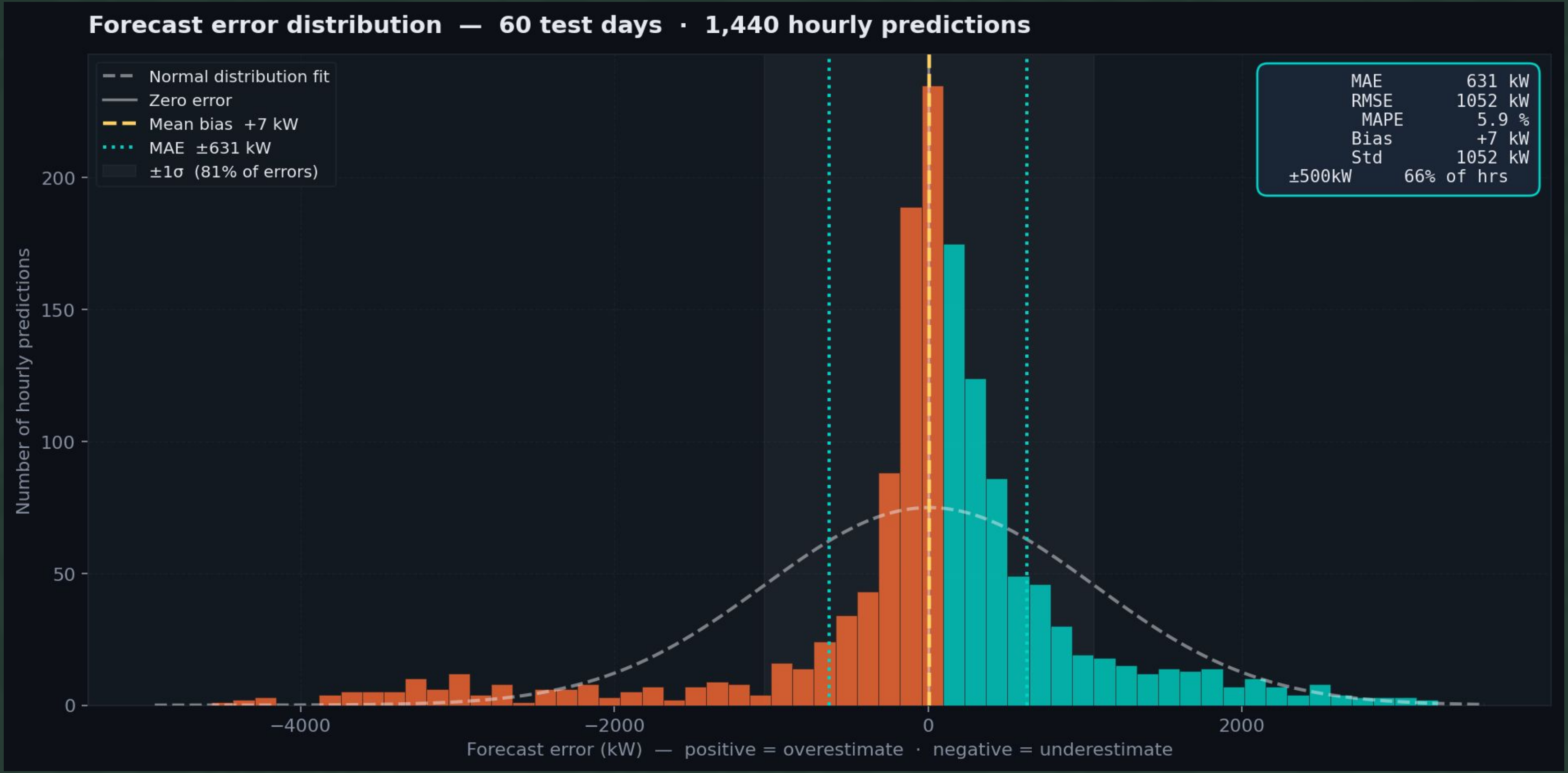
Results

Weekday Vs Weekend

Results

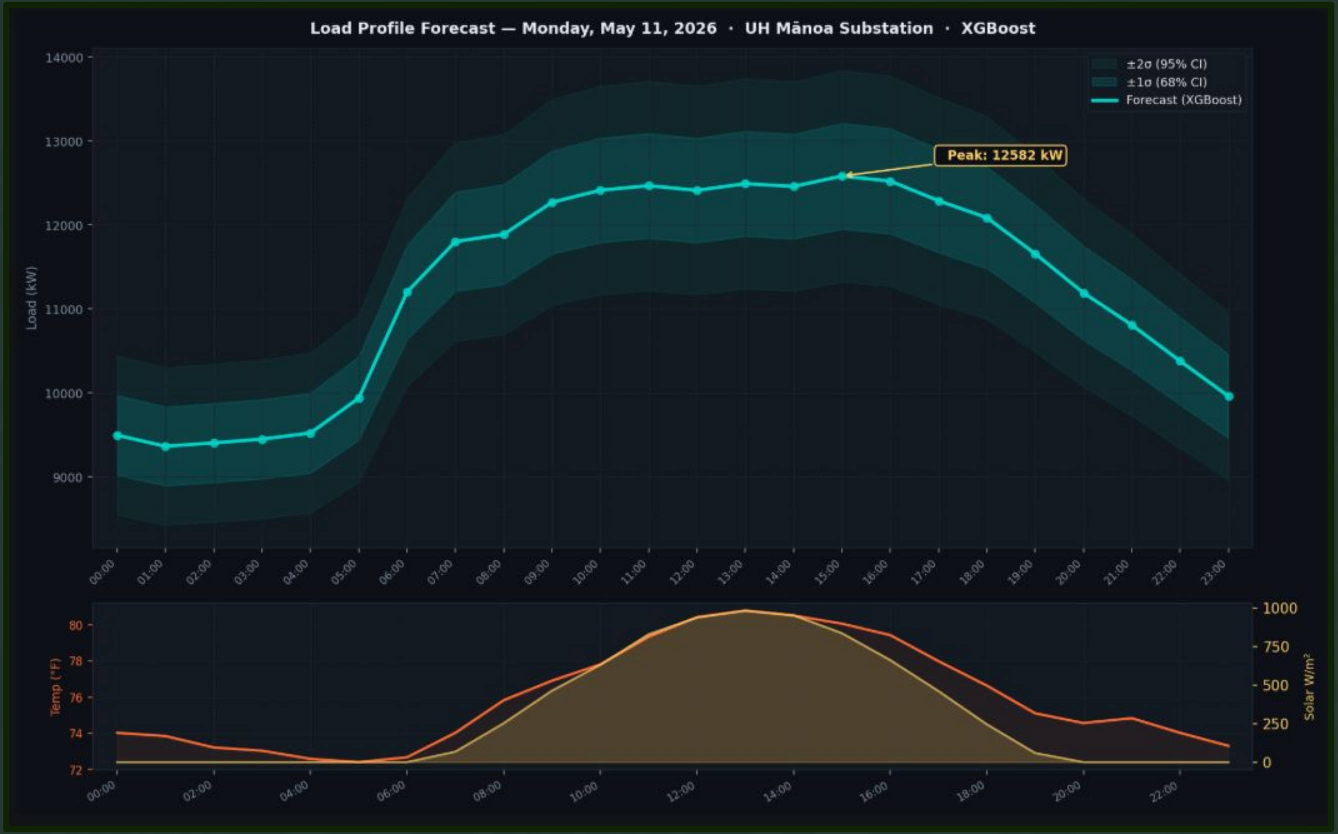
Forecast Days

Results



Forecast error distribution — 60 test days · 1,440 hourly predictions
Normal distribution fit
Zero error
Mean bias +7 kW
MAE ±631 kW
±1σ (81% of errors)
MAE 631 kW
RMSE 1052 kW
MAPE 5.9 %
Bias +7 kW
Std 1052 kW
±500kW 66% of hrs
Number of hourly predictions
0
50
100
150
200
−4000
−2000
0
2000
Forecast error (kW) — positive = overestimate · negative = underestimate

Results



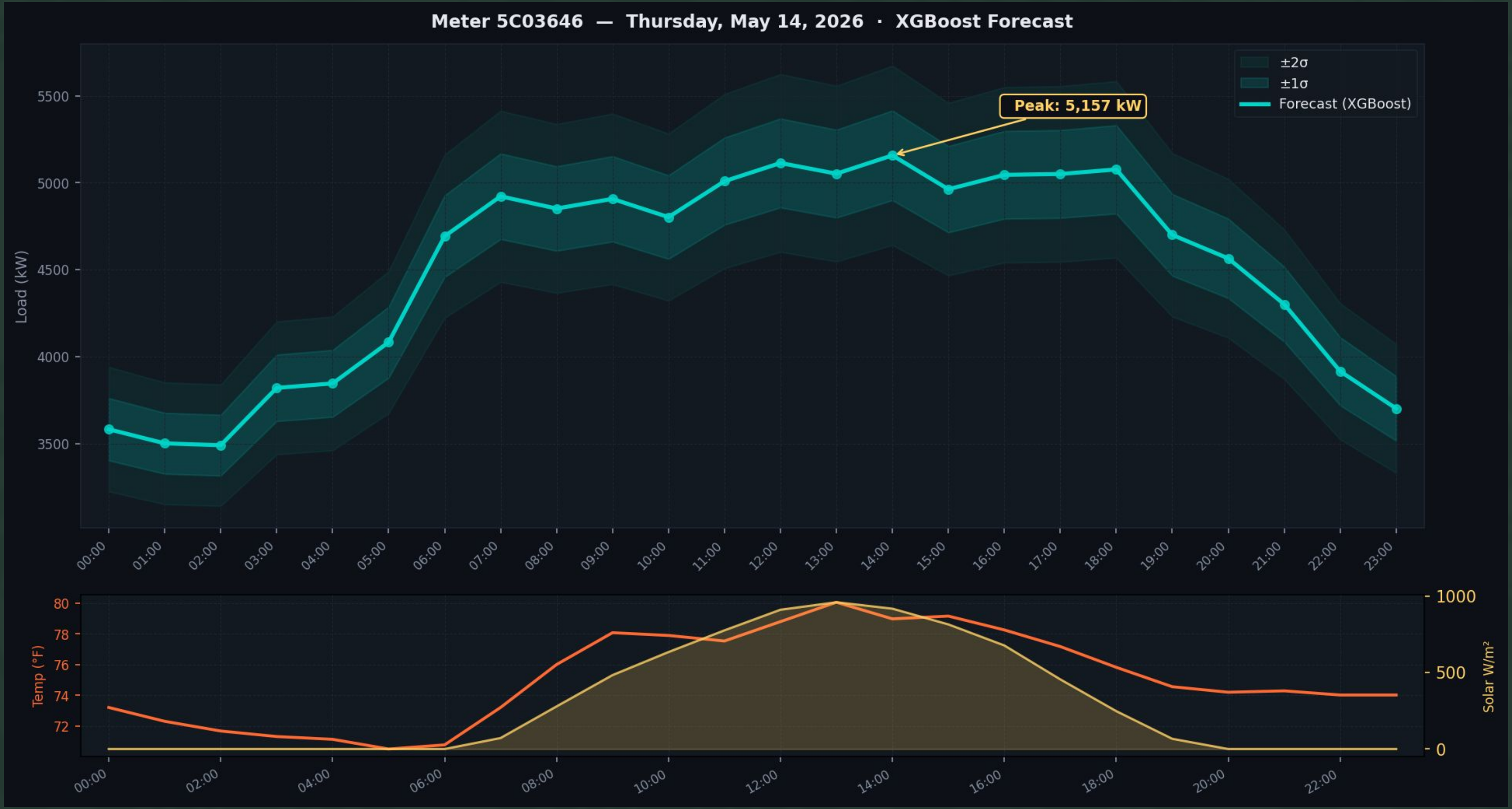
Forecast

Per-Meter Forecasting

Meter 1

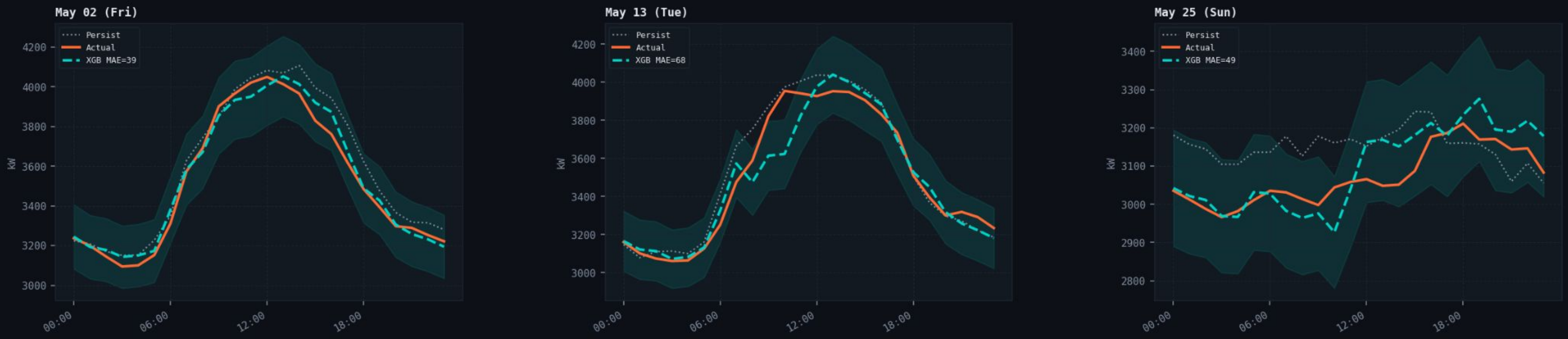
May 02 (Fri)
Persist
Actual
XGB MAE=276
May 13 (Tue)
Persist
Actual
XGB MAE=234
May 25 (Sun)
Persist
Actual
XGB MAE=582
kW
00:00
06:00
12:00
18:00

Meter 1

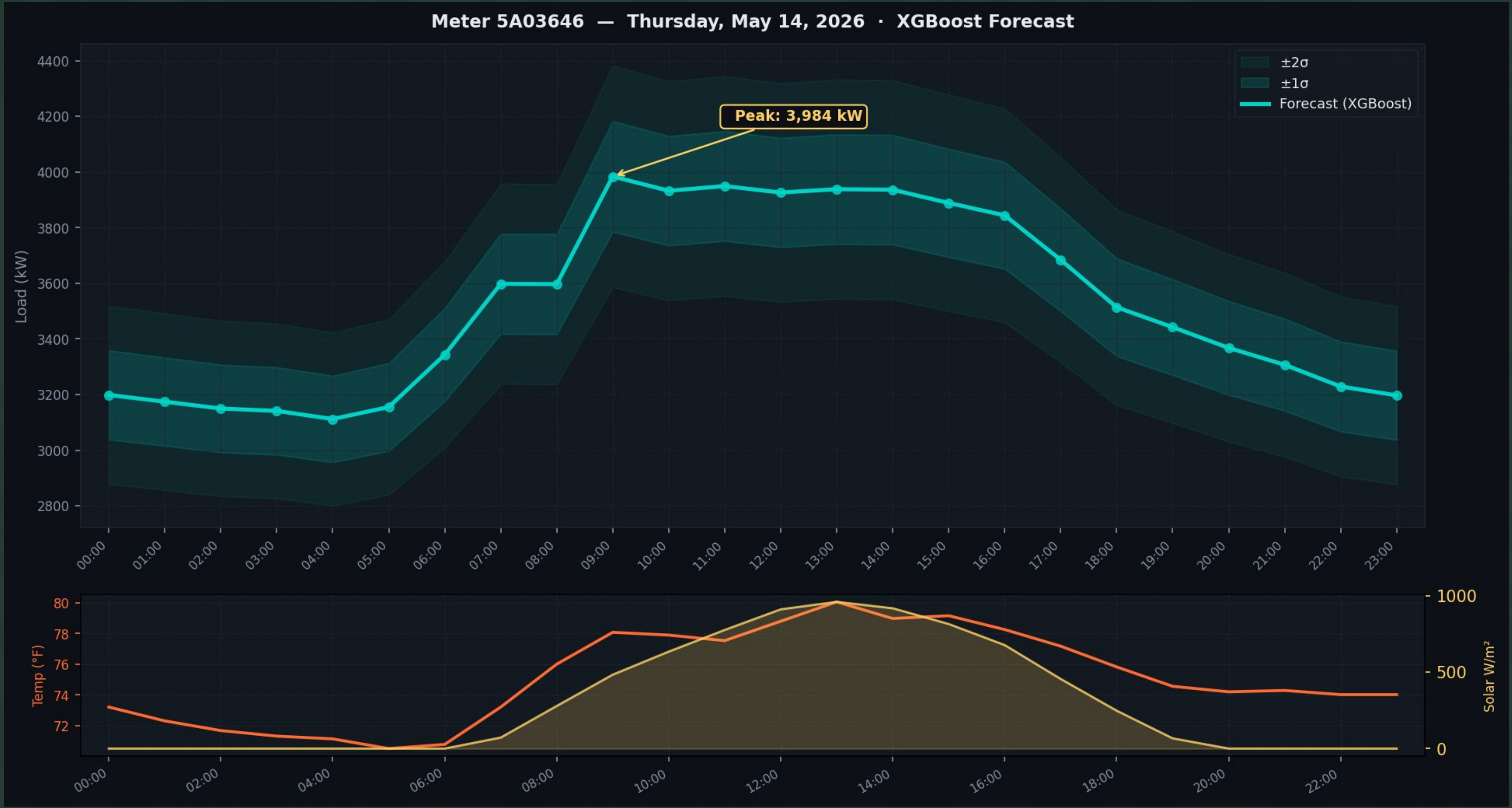


Meter 5C03646 — Thursday, May 14, 2026 · XGBoost Forecast
±2σ
±1σ
Forecast (XGBoost)
Peak: 5,157 kW
Load (kW)
5500
5000
4500
4000
3500
00:00
01:00
02:00
03:00
04:00
05:00
06:00
07:00
08:00
09:00
10:00
11:00
12:00
13:00
14:00
15:00
16:00
17:00
18:00
19:00
20:00
21:00
22:00
23:00
Temp (°F)
80
78
76
74
72
Solar W/m²
1000
500
0

Meter 2


May 02 (Fri)
Persist
Actual
XGB MAE=39
May 13 (Tue)
Persist
Actual
XGB MAE=68
May 25 (Sun)
Persist
Actual
XGB MAE=49
kW
00:00
06:00
12:00
18:00

Meter 2

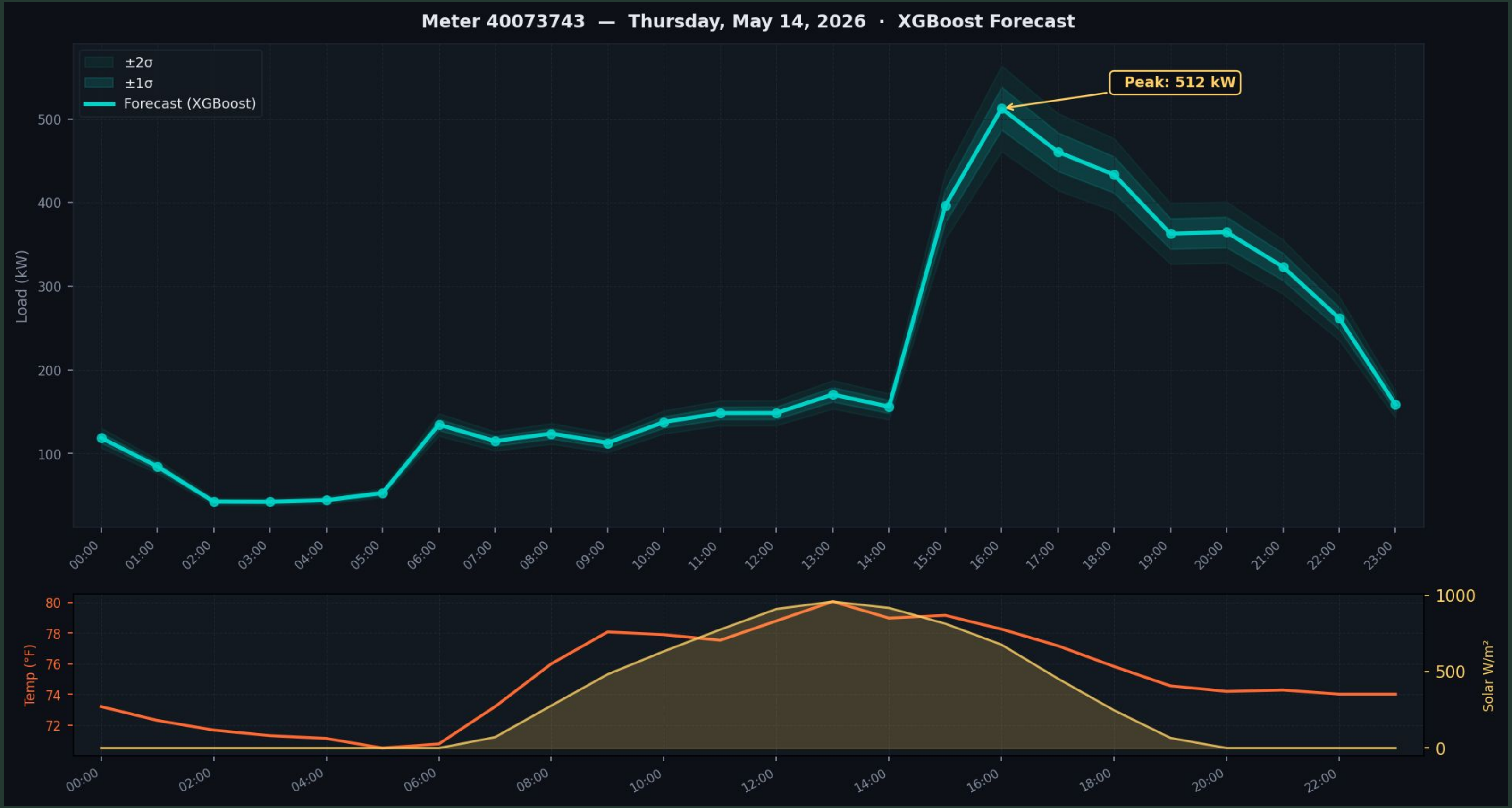


Meter 5A03646 — Thursday, May 14, 2026 · XGBoost Forecast
±2σ
±1σ
Forecast (XGBoost)
Peak: 3,984 kW
Load (kW)
4400
4200
4000
3800
3600
3400
3200
3000
2800
00:00
01:00
02:00
03:00
04:00
05:00
06:00
07:00
08:00
09:00
10:00
11:00
12:00
13:00
14:00
15:00
16:00
17:00
18:00
19:00
20:00
21:00
22:00
23:00
Temp (°F)
80
78
76
74
72
Solar W/m²
1000
500
0
00:00
02:00
04:00
06:00
08:00
10:00
12:00
14:00
16:00
18:00
20:00
22:00

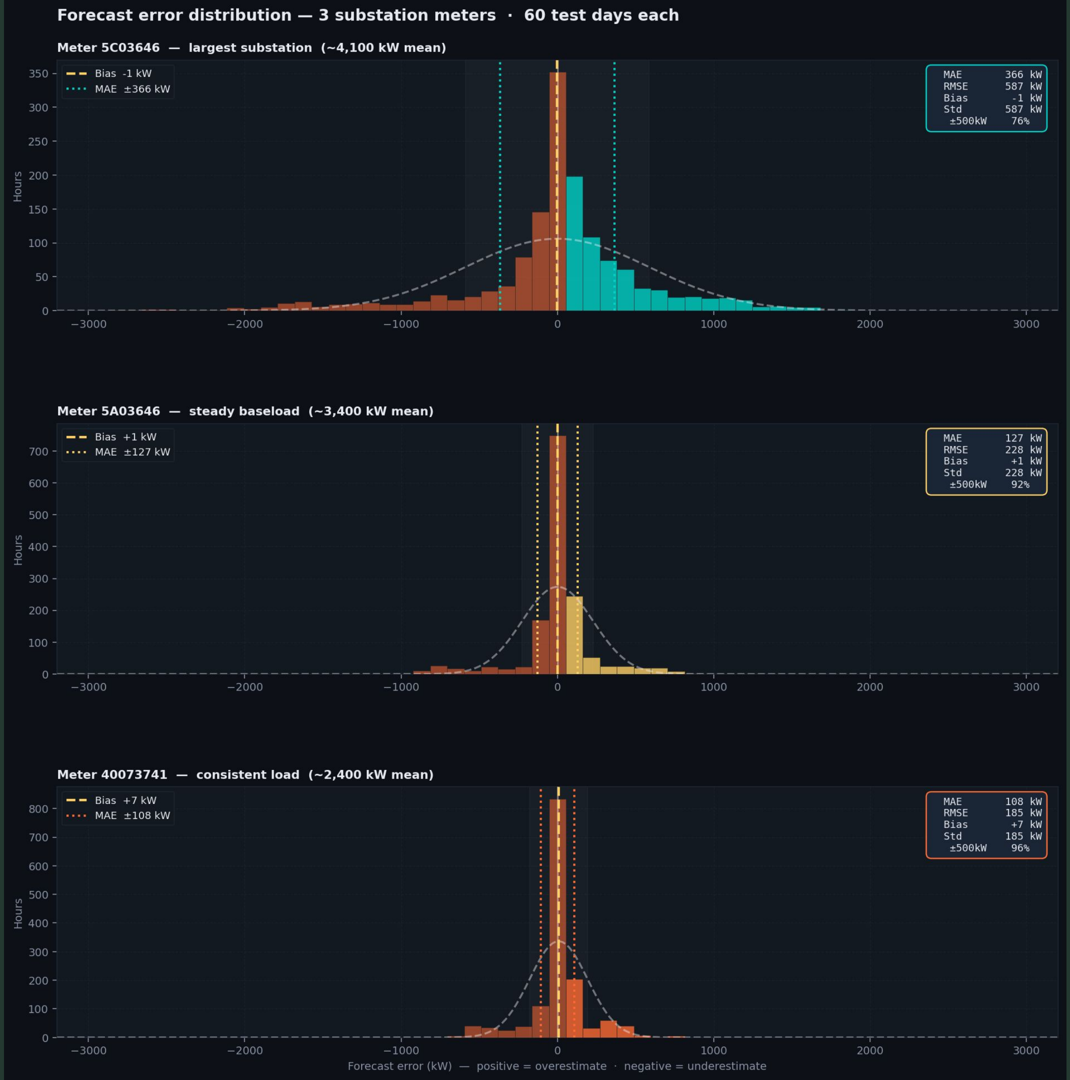
Meter 3


May 02 (Fri)
Persist
Actual
XGB MAE=107
kW
700
600
500
400
300
200
100
00:00
06:00
12:00
18:00
May 13 (Tue)
Persist
Actual
XGB MAE=137
kW
700
600
500
400
300
200
100
00:00
06:00
12:00
18:00
May 25 (Sun)
Persist
Actual
XGB MAE=347
kW
800
700
600
500
400
300
200
100
0
00:00
06:00
12:00
18:00

Meter 3



Meter 40073743 — Thursday, May 14, 2026 · XGBoost Forecast
±2σ
±1σ
Forecast (XGBoost)
Peak: 512 kW
Load (kW)
500
400
300
200
100
00:00
01:00
02:00
03:00
04:00
05:00
06:00
07:00
08:00
09:00
10:00
11:00
12:00
13:00
14:00
15:00
16:00
17:00
18:00
19:00
20:00
21:00
22:00
23:00
Temp (°F)
80
78
76
74
72
Solar W/m²
1000
500
0
00:00
02:00
04:00
06:00
08:00
10:00
12:00
14:00
16:00
18:00
20:00
22:00


Forecast error distribution — 3 substation meters · 60 test days each
Meter 5C03646 — largest substation (~4,100 kW mean)
Bias -1 kW
MAE ±366 kW
MAE 366 kW
RMSE 587 kW
Bias -1 kW
Std 587 kW
±500kW 76%
Meter 5A03646 — steady baseload (~3,400 kW mean)
Bias +1 kW
MAE ±127 kW
MAE 127 kW
RMSE 228 kW
Bias +1 kW
Std 228 kW
±500kW 92%
Meter 40073741 — consistent load (~2,400 kW mean)
Bias +7 kW
MAE ±108 kW
MAE 108 kW
RMSE 185 kW
Bias +7 kW
Std 185 kW
±500kW 96%
Hours
Forecast error (kW) — positive = overestimate · negative = underestimate

Mahalo!